Catalysis

Chiral Oxazolinyl Hydroxyl Ferrocene Catalysts: Synthesis and Applications in Asymmetric Reactions

Di Xu, Jun Zhang, and Li Dai*[a]

Chiral oxazolinyl hydroxyl ferrocenes have been demonstrated as representative catalysts in enantioselective nucleophilic addition reactions in the last few decades. They have received much interest from chemists because of not only the numerous substitution patterns at their sandwich backbones, but also their superior catalytic activity in asymmetric transformations. Recent research has revealed that the planar chirality of these catalysts influences the stereochemistry of asymmetric products and the application scope in enantioselective reactions. Herein, we provide an intensive survey of chiral oxazolinyl hydroxyl ferrocene compounds, including the structures of chiral compounds already published, designed synthetic approaches for ferrocenes with different patterns, and their utilization in the enantioselective preparation of secondary alcohols using organozinc reagents.

1. Introduction

Based on seminal work in 1951, *ferrocene* has been demonstrated as a landmark in the field of organometallic chemistry. Ferrocene is symbolized by its sandwich structure and fascinated by its properties such as structural rigidity, thermal stability, oxygen and moisture tolerance, which captured attention of researchers in medicinal chemistry, materials science and organic synthesis. Moreover, the aromatic cyclopentadienyl (Cp) ring is easily functionalized by electrophilic substitution. For instance, thousands of ligands, with central, axial, and spiro chirality, featuring various stereogenic elements embedded in ferrocene frameworks. Most importantly, an interesting structural feature of ferrocene-based derivatives is planar chirality, which contributed by introducing two or more different groups on single Cp ring. Therefore, significant improvements on preparation of planar chiral ferrocene compounds have been developed by several elegant synthetic strategies. In many cases, the absolute configuration of planar chirality of ferrocene-based ligands determines diastereo- and enantioselectivity of corresponding chiral products in transition-metal catalyzed transformations. Nowadays, chiral ferrocene-type compounds have been recognized as essential ligands/catalysts in both academic studies and industrial applications. Several excellent book chapters and reviews have been published in accord with the significance chiral ferrocenes.[1–7]

Among the explored ferrocenyl ligands, oxazoline-substituted ferrocenes in particular have received considerable attention over the past several years. Chiral oxazolines have been demonstrated to have strong coordination ability with transition metals. Chiral bisoxazolines (Box) with symmetrical oxazoline. units have been employed as ligands in a wide range of transition-metal-catalyzed asymmetric reactions.[8,9] These groups have also appeared in several privileged structures, for example, phosphino-oxazoline (PHOX) ligands and chiral spiro ligands.[10] Additionally, oxazoline fragments have been used as auxiliary groups in the diastereoselective synthesis of planar chiral ferrocenes.[11] Because of the steric bulkiness of both the chiral substituents on the oxazoline and the cyclopentadienyl (Cp) rings, diastereomeric ratios for lithiated ferrocene intermediates in excess of 500:1 can be obtained.[12] Consequently, planar chiral oxazolinyl ferrocenes have been furnished by installing various counter chelating groups, such as groups with phosphorus,[13–15] sulfur,[16] nitrogen,[17] and even carbene (*N*-heterocyclic carbene)[18,19] moieties.

In contrast to the aforementioned oxazolinyl ferrocenes, chiral hydroxyl ferrocene catalysts have received only limited interest from researchers during the same period. However, chiral *N*,*O*-bidentate derivatives have been employed as homogeneous catalysts in zinc-mediated enantioselective additions to carbonyl groups.[20,21] In 2010, ferrocene-based catalysts that have been applied to this type of addition were summarized by Bolm and coworkers.[22] On the subject of hydroxyl oxazolinyl catalysts, the premier work was the synthesis and application of **1** and **2** by the Williams group in 1994 (Figure 1).[23] Soon after, a seminal example of a chiral oxazolinyl hydroxyl ferrocene was reported by the Bolm group.[24] Since then, the number of oxazolinyl hydroxyl ferrocene catalysts has increased gradually, owing to the great efforts of various research groups. Thus, the objective of this work is to present a specialized overview of these ferrocenes, detailing their synthetic methods and catalytic performances in asymmetric reactions, especially the recent developments achieved prior to the end of 2019.

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1a: R^1 = Me, R^2 = Ph, R^3 = H;
1b: R^1 = Ph, R^2 = Ph, R^3 = H;
1c: R^1 = 2-thiophenyl, R^2 = Ph, R^3 = H;
1d: R^1 = Me, R^2 = H, R^3 = Me;
1e: R^1 = Me, R^2 = H, R^3 = Ph;

2a: R = Me;
2b: R = Ph;

Figure 1. Chiral oxazolinyl hydroxyl catalysts.

2. Structures of chiral oxazolinyl hydroxyl ferrocenes

The chiral oxazolinyl hydroxyl ferrocenes that have already been published can be classified into three categories, according to the substitution pattern in their ferrocene cores.

(a) Planar chiral oxazolinyl hydroxyl ferrocenes

Planar chiral compounds have been identified as the main *N*,*O*-bidentate ferrocenes. Bolm has reported the synthesis of planar chiral catalysts **3a**[24] and **3b**, as well as **4** with only planar chirality.[25] In his studies, both (*S*, *Rp*)- and (*S*, *Sp*)-**3a** were synthesized. Butenschön enlarged the structures of the side chains containing the hydroxyl groups in **3c**–**e** by using ferrocenyl ketones as electrophiles to quench *ortho*-lithiated intermediates.[26] Guiry synthesized *gem*-disubstituted *N*,*O*-ferrocenes (*Sp*)-**5** and (*Rp*)-**5**.[27] Very recently, motivated by our previous work,[28,29] we successfully prepared a range of *N*,*O*-ferrocenes, **6**, containing silyl ethers.[30] Bolm presented an example of an alternative hydroxyl moiety. Silanol groups were attached to the Cp ring, affording planar chiral organosilanol ferrocenes **7**.[31] Planar chiral C_2-symmetric *N*,*O*-ferrocenes were discussed by Ikeda and coworkers. Tetrasubstituted ferrocenes **8** were prepared from C_2-symmetric bisoxazoline ferrocene reagents.[32]

Based on the concept of catalyst immobilization, polymer-supported catalysts were later reported. Bolm and coworkers synthesized MeO-polyethylene glycol (PEG)-supported (*S*, *Rp*)-**9** and discussed its application in enantioselective addition to aldehydes.[33] Recently, Zhao and Liu reported compound **10**, a different type of planar chiral ferrocene in which the polymer moiety is attached to the chiral oxazoline ring.[34]

(b) 1,1′-Substituted oxazolinyl hydroxyl ferrocenes

Some examples of 1,1′-substituted *N*,*O*-ferrocenes **11** were reported by Hou and coworkers. In this type of compound, the oxazoline moiety and side chain containing the hydroxyl group are attached to the two different Cp rings.[35,36]

(c) Mono-substituted oxazolinyl hydroxyl ferrocenes

Bonini and Zwanenburg reported mono-substituted **12a** and **12b** and C_2-symmetric **13a**, which were prepared through the

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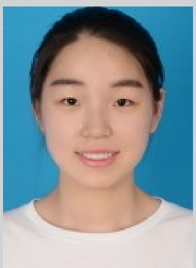

Jun Zhang became a student majoring in applied chemistry at North China University of Water Resources and Electric Power (NCWU) in 2017. Since 2018, she began his BSc study under the direction of Dr. Li Dai. She is currently worked on the synthesis and application of silyl-modified oxazolinyl hydroxyl ferrocene catalysts.

Li Dai was born in Changchun, China in 1986. He graduated from Beijing Institute of Technology (BIT) obtaining a BSc in fine chemical engineering in 2009. He received his Ph.D. degree in applied chemistry from BIT under the supervision of Prof. Zhi-Ming Zhou in 2015. After a postdoctoral stage on ionic liquids under the guidance Prof. Suojiang Zhang at the Institute of Process Engineering, Chinese Academy of Sciences (IPE, CAS), he appointed to a lectureship in the North China University of Water Resources and Electric Power (NCWU). His current research interests focus on the asymmetric synthesis and catalysts, as well as on the functionalized ionic liquids.

Reviews
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ring expansion of aziridines.[37] Richards presented chiral *N,O*-ferrocenes with a variety of backbones, including ferrocene-based **12 c** and **12 d**.[38] C_2-symmetric ferrocene derivatives have also been discussed, and the corresponding C_2-symmetric bisoxazolines **13 b**–**e** were prepared by Zhang and coworkers by reducing oxazoline esters.[39]

3. Synthesis of chiral oxazolinyl hydroxyl ferrocenes

3.1 Planar chiral oxazolinyl hydroxyl ferrocenes

3.1.1 Conventional planar chiral N,O*-ferrocene catalysts*

As shown in Figure 2, ferrocene catalysts with planar chirality have received more attention from chemists. According to published synthetic processes, most planar chiral ferrocene derivatives were obtained through a diastereoselective directed *ortho*-metalation strategy. Chiral oxazolines have been employed as *N*-based directing groups for metalation, as indicated by Sammakia's model.[40] With alkyllithium species, repulsive interactions with both ferrocene and the chiral substituent group on the oxazoline ring should be avoided during *ortho*-metalation. Therefore, planar chiral ferrocenes can be obtained easily from *ortho*-lithiated oxazolinyl ferrocenes with high diastereoselectivities (mostly above 99% diastereomeric ratio) by quenching with an appropriate electrophile. Ferrocenes (*S*, *Rp*)-**3**, (*S*, *Rp*)-**5**, (*S*, *Sp*)-**6**, and (*S*, *Sp*)-**7** were synthesized successfully following the aforementioned strategy (Scheme 1).

However, *ortho*-metalation of achiral 4,4-dimethyloxazolinyl ferrocene afforded only racemic **4**. Hence, Bolm prepared (*Sp*)-iodooxazolinyl ferrocene for the synthesis of (*Rp*)-**4** (Scheme 2).[25]

3.1.2 Reversed planar chiral N,O*-ferrocenes*

The absolute planar chiral configurations of ferrocenyl catalysts dramatically impact the stereoselectivity of enantioselective reactions. As a result, several methods have been proposed for the construction of diastereomeric ferrocenes.

In general, the "silicon trick" is a common strategy for disfavored planar chirality.[41] Bolm constructed the opposite planar chiral ferrocenes by using a temporary group (trimeth-

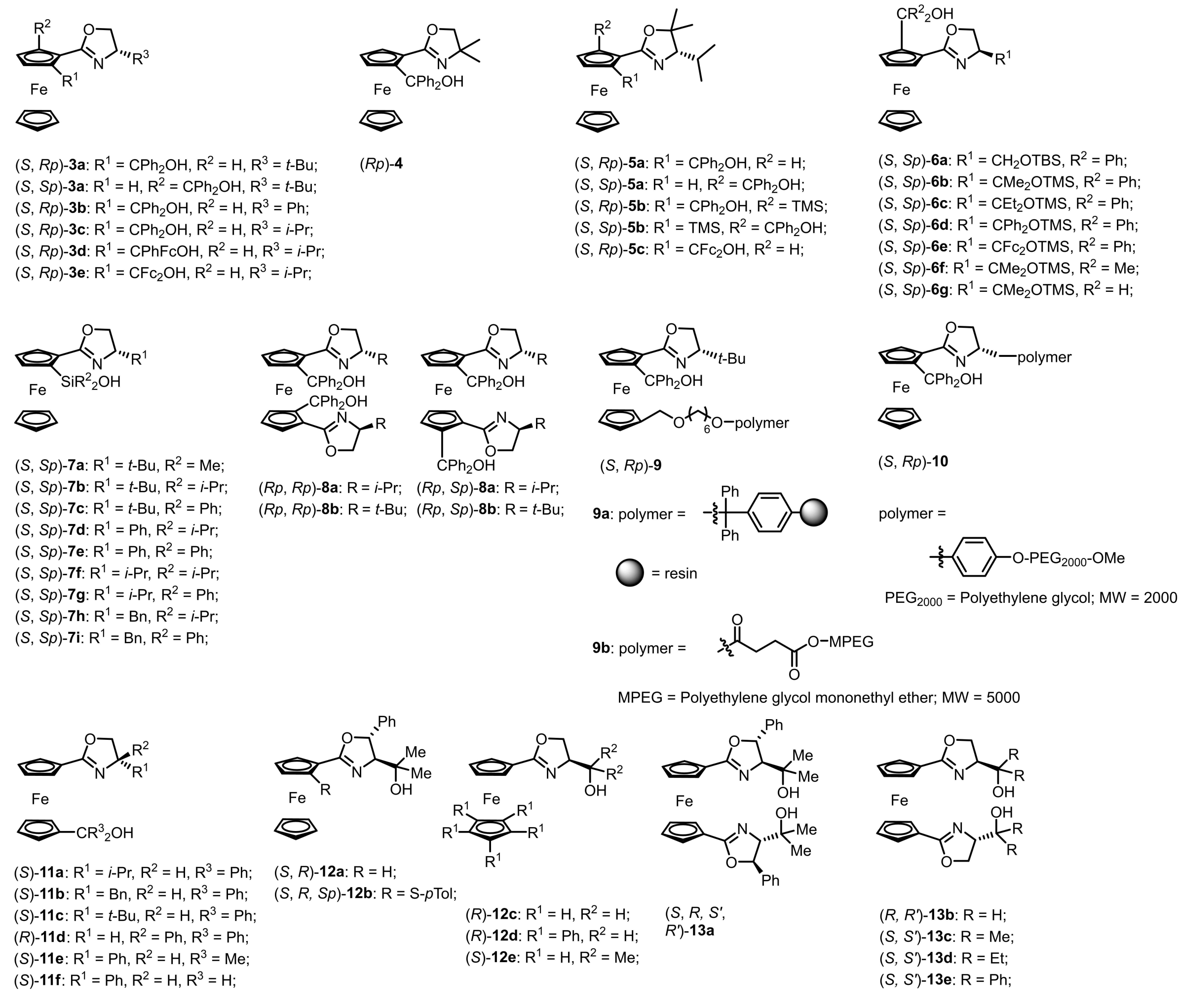
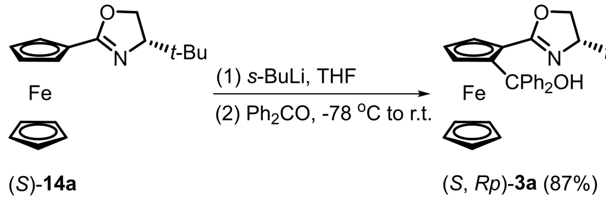
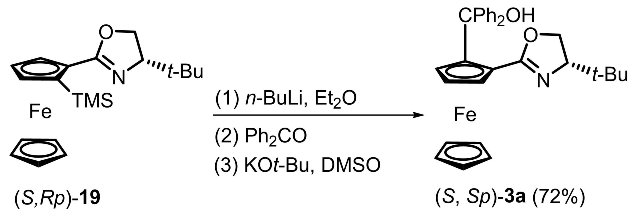


Figure 2. Chiral oxazolinyl hydroxyl ferrocenes.



Scheme 1. Preparation of conventional planar chiral oxazolinyl hydroxyl ferrocene (*S*, *Rp*)-**3 a**.



Scheme 3. Preparation of oxazolinyl hydroxyl ferrocene (*S*, *Sp*)-**3 a** with a temporary TMS group.

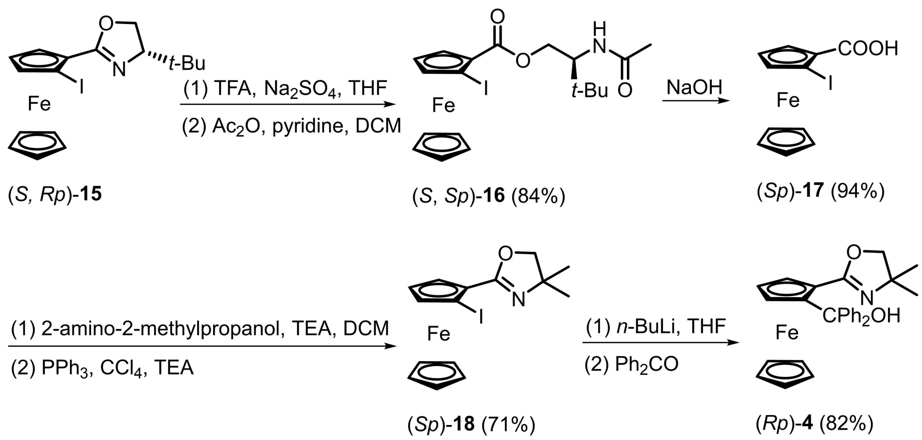
ylsilyl, TMS) that blocked the preferred position. An additional *ortho*-lithiation-quenching-desilylation sequence afforded desired ferrocene (*S*, *Sp*)-**3 a** (Scheme 3).[25]

The diastereoselectivity of directed *ortho*-metalation could be increased further by chelating agents. For example, in the presence of tetramethylethylenediamine (TMEDA), the directed *ortho*-metalation provided diastereomeric ratios higher than 500 : 1.[12] In 2013, Arnott reported that planar chirality changes significantly in the presence of diglyme ligands.[42] Guiry extended this methodology to the synthesis of (*S*, *Rp*)-**22**. However, **22** was obtained with only 37.5% diastereomeric excess (de), so further preparation of reversed ferrocene catalysts has been limited (Scheme 4).[27]

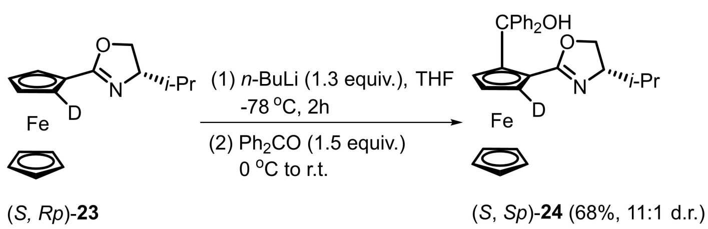
Recently, Richards applied deuterium as an "invisible" blocking group in the preparation of reversed planar chiral oxazoline ferrocenes (Scheme 5).[43] Deuterated ferrocene (*S*, *Rp*)-**23** was then converted into a range of reversed planar chiral ferrocene compounds with good diastereomeric ratios. The diastereoselectivity was determined by the kinetic isotope effect. As a result, (*S*, *Sp*)-oxazolinyl hydroxyl ferrocene **24** was obtained in 68% yield with a d.r. of 11 : 1.

3.1.3 Planar chiral oxazolinyl silanol ferrocenes

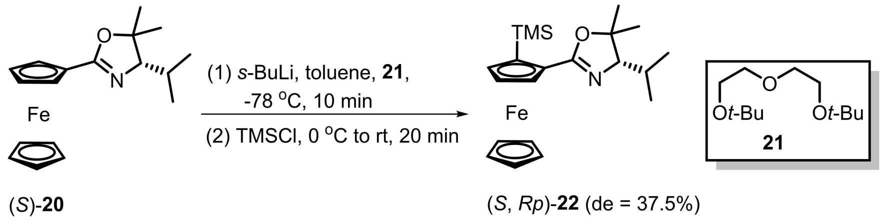
In 2005, organosilanol fragments were employed as chelating groups by Bolm and coworkers.[31] After the standard diastereoselective *ortho*-lithiation step, various chlorosilanes were used as electrophiles to provide 1,2-disubstituted ferrocenes **25**. Then, organosilanols 7 were obtained via silyl group oxidation catalyzed by $[IrCl(C_8H_{12})]_2$ in air (Scheme 6).



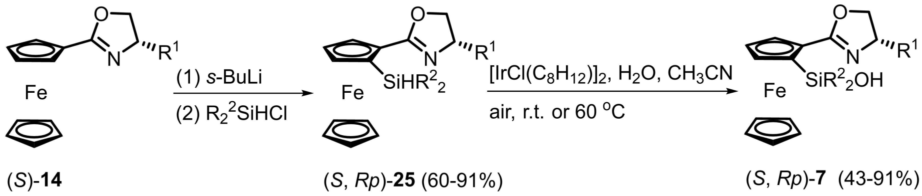
Scheme 2. Preparation of planar chiral oxazolinyl hydroxyl ferrocene (*Rp*)-**4**.



Scheme 5. Deuterium-directed synthesis of oxazolinyl hydroxyl ferrocene (*S*, *Sp*)-**24**.



Scheme 4. Diastereoselective synthesis of (*S*, *Rp*)-**22** with **21** as a chelating agent.



Scheme 6. Preparation of planar chiral oxazolinyl silanol ferrocenes **7**.

3.1.4 C$_2$-Symmetric oxazolinyl hydroxyl ferrocenes

C$_2$-Symmetric oxazolinyl hydroxyl ferrocenes were reported by Ikeda and coworkers in 2000.[32] The synthesis began with *C*$_2$-symmetric oxazolinyl ferrocenes **26**, which were treated with 2.6 equiv. of *s*-BuLi and TMEDA. After the reaction was quenched with benzophenone, desired *C*$_2$-symmetric hydroxyl ferrocenes (*Rp*, *Rp*)-**8** were obtained in yields higher than 60% (Scheme 7). Interestingly, *C*$_1$-symmetric (*Rp*, *Sp*)-**8** was obtained simultaneously through this lithiation procedure, mainly because of the steric repulsion of the substituents on the oxazoline ring of **26** during the dilithiation step.

3.1.5 Polymer-supported oxazolinyl hydroxyl ferrocenes

Polymer groups were introduced into the ferrocene frameworks for catalyst recycling. In Bolm's procedure, an alkyl chain was attached directly to a Cp ring (Scheme 8). Then, the hydroxyl group was reacted with two different polymers, affording ferrocenes (*S*, *Rp*)-**9**.[33]

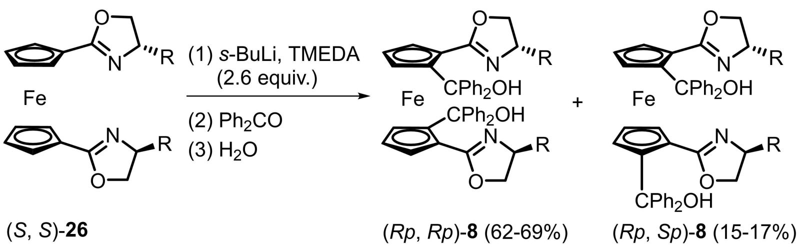
Zhao and Liu disclosed different polymer-supported compounds in 2016 (Scheme 9).[34] Planar chiral oxazolinyl hydroxyl ferrocene was prepared from *tert*-butyldimethylsilyl (TBS)-protected ʟ-tyrosine derivative **30**. After the protecting group was removed, the hydroxyl group was reacted with a polymer mesylate to afford the target compound, (*S*, *Rp*)-**10**.

3.2 1,1′-Substituted oxazolinyl hydroxyl ferrocenes

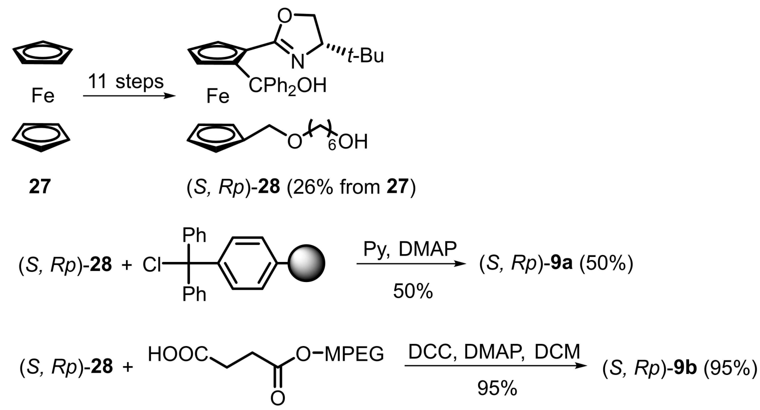
Debromolithiation is a common procedure for introducing electrophiles into a ferrocene skeleton, especially for unsymmetrical 1,1′-substituted ferrocene compounds. Hou reported 1,1′-substituted oxazolinyl hydroxyl ferrocenes **11 a**–**d** from 1-bromo-1′-oxazolinyl ferrocenes in 73–93% yields through halogen-lithium exchange followed by quenching with benzophenone (Scheme 10, equation 1).[35] Then, the halogen group on the Cp ring was applied in a similar process for the preparation of 1,1′-substituted oxazolinyl ferrocene carbinols **11 e** and **11 f** using different electrophiles (Scheme 10, equations 2 and 3).[36]

3.3 Mono-substituted C$_2$-symmetric oxazolinyl hydroxyl ferrocenes

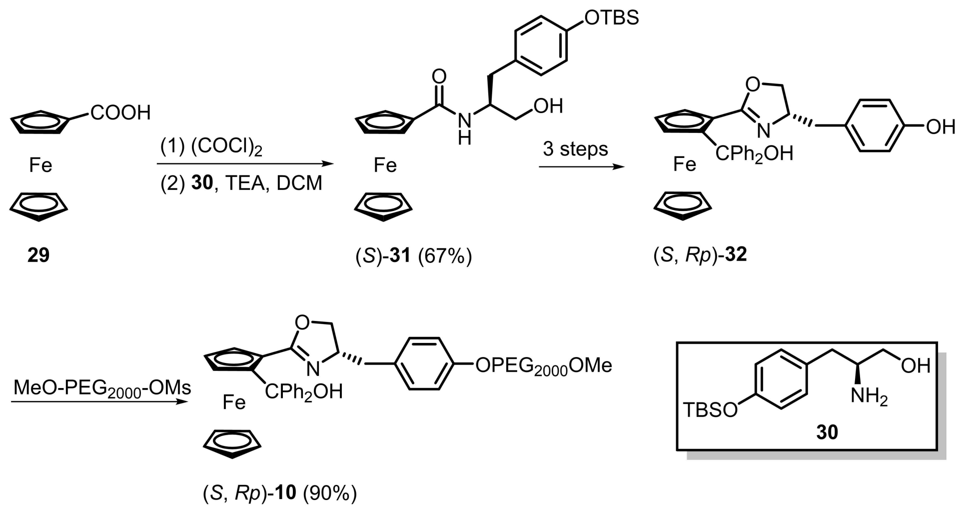
Mono-substituted oxazolinyl hydroxyl ferrocenes are typically produced from oxazoline 4-carboxylic esters. According to the literature, two different materials can be applied to the synthesis of oxazoline esters. In 2003, Bonini and Zwanenburg



Scheme 7. One-pot preparation of *C*$_2$-symmetric (*Rp*, *Rp*)-**8** and *C*$_1$-symmetric (*Rp*, *Sp*)-**8**.



Scheme 8. Preparation of polymer-supported ferrocenes (*S*, *Rp*)-**9**.



Scheme 9. Preparation of MeO-PEG-supported ferrocene (*S*, *Rp*)-**10**.

used *N*-ferrocenoyl-aziridine-2-carboxylic esters as reactants for the preparation of oxazoline 4-carboxylic esters in the presence of NaI in refluxing acetone. The resulting oxazolines were then treated with Grignard reagents to afford ferrocenes **12a**, **12b**, and **13a** (Scheme 11).[37]

β-Hydroxyl amides were independently identified as alternative reactants for the synthesis of oxazoline 4-carboxylic esters by the Richards and Zhang groups. Richards reported the cyclization of oxazolines in nearly quantitative yields using diethylaminosulfur trifluoride (DAST) (Scheme 12). Moreover, this efficient cyclization procedure was also applied in the synthesis of mono-substituted ferrocenes **12c** and **12d**.[38]

Zhang prepared oxazoline 4-carboxylic ester by using the Burgess reagent [(methoxycarbonylsulfamoyl)-triethylammonium hydroxide inner salt] from C_2-symmetric ferrocene diamine **40**. The resulting C_2-symmetric bisoxazoline derivative **41** were obtained in 70% yield (Scheme 13).[39]

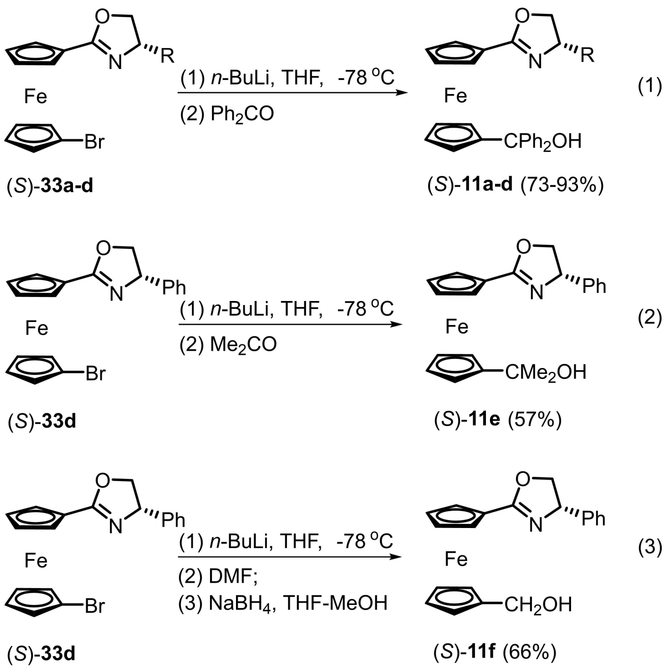
Utilizing the synthetic approaches above, more than 50 ferrocenes have been prepared. There absolute configurations, synthetic yields (mostly from corresponding oxazolinyl ferrocene) and diastereoselectivities are listed in Table 1.

4. Applications of ferrocene catalysts in asymmetric reactions

4.1 Asymmetric addition of dialkylzinc to aldehydes

Because of the biological activity of secondary alcohols, the addition of diethylzinc to aldehydes is the most frequently employed reaction in the screening of chiral *N*,*O*-ferrocene catalysts. A selection of significant contributions involving ferrocene catalysts is presented in Table 2.

In 1997, Bolm and coworkers reported the first planar chiral ferrocene catalyst, (*S*, *Rp*)-**3a**, to be applied to asymmetric additions. Using this catalyst, moderate to good enantiomeric excesses (ee) were achieved for aromatic and aliphatic alcohols in additions to aldehydes.[24] A further study on planar chirality showed that appropriate steric repulsion for higher enantioselectivity could be attained through cooperative chirality.[25]



Scheme 10. Preparation of unsymmetric 1,1′-substituted oxazolinyl hydroxyl ferrocenes **11**.

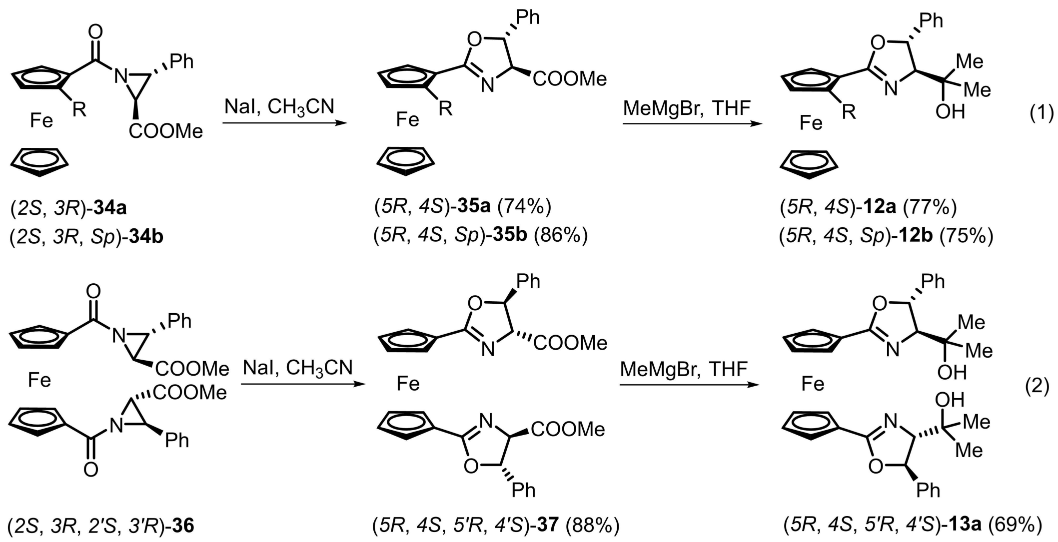
Moderate enantioselectivity in a reaction catalyzed by (*Rp*)-**4**, which has only planar chirality, indicated that central chirality was the dominant factor in the stereochemistry.

Attempts to use triferrocenylmethanol (*S*, *Rp*)-**3e** in asymmetric addition were made by the Butenschön group.[26] The bulkier group suppressed the catalytic activity but had a beneficial effect on the enantioselectivity (97% ee), presumably owing to the transition state being more stable because of the ferrocene.

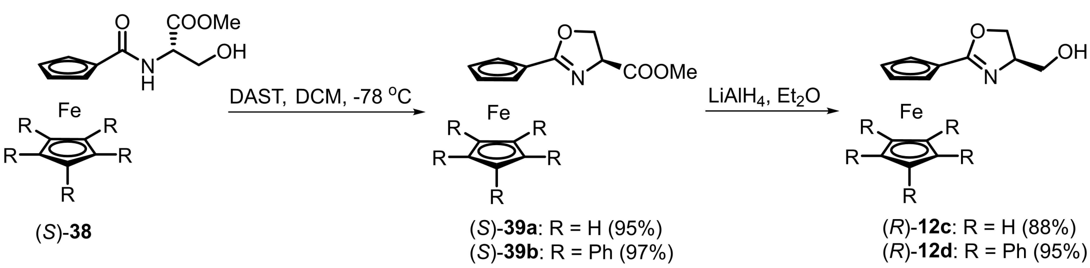
A series of *gem*-disubstituted *N*,*O*-ferrocenes, **5**, were tested by Guiry.[27] Because the structure and configuration of the central chiral group in the catalysts was the same, the planar chirality was demonstrated to be a prominent factor in the enantioselectivity. Excellent enantioselectivities were achieved in the additions to aromatic aldehydes using (*S*, *Rp*)-**5c**, which featured a ferrocene fragment attached to the methanol unit. Interestingly, in a related study, high ee values were obtained in additions to aliphatic aldehydes using (*S*, *Rp*)-**5a**.

Ikeda and coworkers studied planar chiral tetrasubstituted oxazolinyl ferrocenes in asymmetric additions to benzaldehyde.[32] (*Rp*, *Sp*)-**8b** was the optimal catalyst, rather than C_2-symmetric (*Rp*, *Rp*)-**8b**, providing 97% ee.

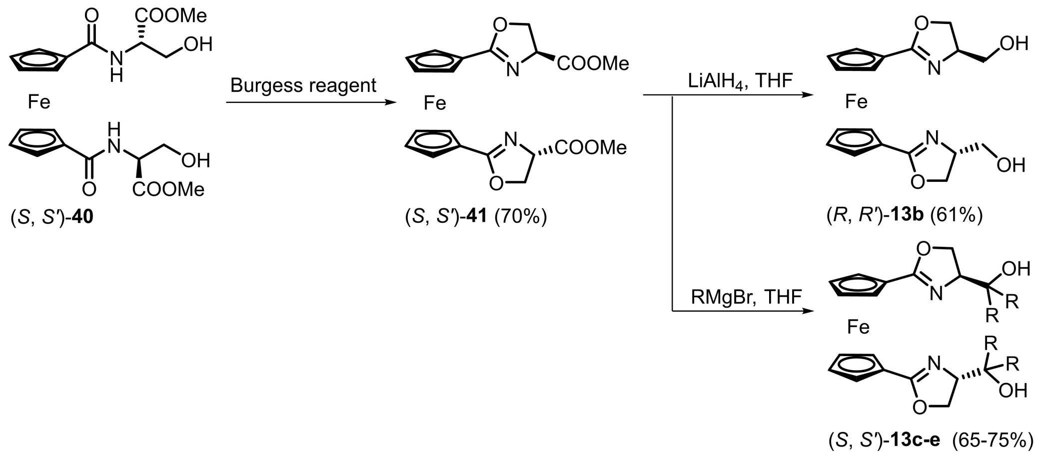
Different substitution patterns on the ferrocene catalysts were also investigated in this asymmetric reaction. 1,1′-



Scheme 11. Preparation of mono-substituted ferrocenes through ring expansion of *N*-ferrocenoyl-aziridines.



Scheme 12. Preparation of mono-substituted oxazolinyl hydroxyl ferrocenes (*R*)-**12c** and (*R*)-**12d**.



Scheme 13. Preparation of C_2-symmetric bisoxazoline hydroxyl ferrocenes **13 b**–**e**.

Substituted ferrocene catalysts (*S*)-**11 a**–**d** were reported by Hou and coworkers.[35] All of the chiral alcohols were obtained in good to high yields but with lower enantioselectivities than those obtained with the analogous 1,2-substituted catalysts.

The pioneering work on mono-substituted oxazolinyl ferrocenes in asymmetric addition was presented by Bonini and Zwanenburg.[37] Moderate results (up to 62% ee) were achieved using ferrocene **13 a**. Soon after, higher enantioselectivity (75% ee) was reported by Richards.[38] In 2007, C_2-symmetric bisoxazolinyl ferrocenes **13 b**–**e** were examined by Zhang and coworkers, and these catalysts gave the highest ee at 87%.[39] It is interesting to note that both Richards and Zhang claimed that increasing the size of the bulky groups on the catalysts resulted in decreased enantioselectivity.

In 1999, Bolm and coworkers reported their study on $ZnMe_2$ addition to benzaldehyde, which investigated the relationship between the chirality of a ferrocene catalyst and the product alcohol (Scheme 14).[44] Definite linearity was found between the enantiomeric excesses of chiral alcohols and ferrocene catalysts (*S*, *Rp*)-**3 a** and (*Rp*)-**4** (Figure 3). In addition, a non-linear relationship was found for (*S*, *Rp*)-**3 a** and (*S*, *Sp*)-**3 a** (Figure 4). This was attributed to the different reaction rates of the two planar chiral diastereomeric ferrocenes.

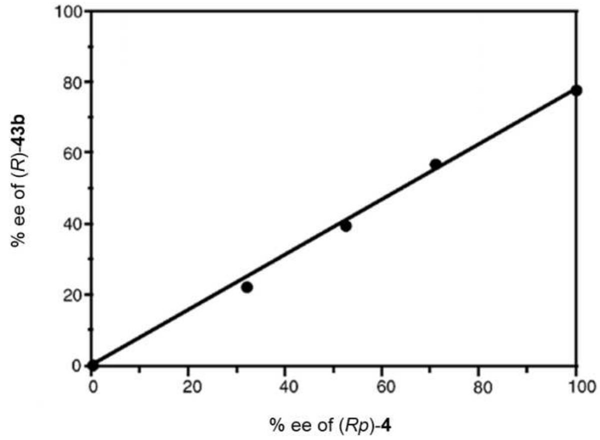
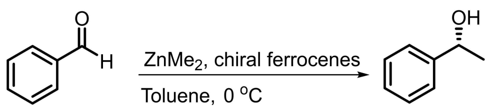


Figure 3. Linear relationship between enantiomeric excesses of (*R*)-**43 b** and (*Rp*)-**4**. Adapted with permission from ref. [44]. Copyright 1999 American Chemical Society.

4.2 Asymmetric phenyl transfer to aldehydes

Phenyl transfer to aldehydes has been described as a practical alternative for the synthesis of chiral diarylmethanols, which are considered to be important precursors for various biologically active compounds. Chemists have devoted efforts to the investigation of aryl sources for this transformation, to improve enantioselectivity and suppress the background reaction, with diethylzinc as a catalyst. Representative results for the enantioselective synthesis of secondary alcohols are listed in Table 3.



Scheme 14. Enantioselective dimethylzinc addition to benzaldehyde **42 a**.

The first example of asymmetric phenyl transfer catalyzed by oxazolinyl hydroxyl ferrocenes was reported by Bolm and coworkers.[45] Under the optimal reaction conditions, using diphenylzinc as a direct aryl source, ferrocenes (*S*, *Rp*)-**3 a** and (*S*, *Rp*)-**3 b** both afforded the desired diarylmethanol in quantitative yield and with 88% ee. Further screening of aromatic aldehydes led to the preparation of a variety of diarylalcohols with high enantioselectivities (up to 96% ee).

A modified aryl source, composed of diphenylzinc and diethylzinc in a 1:2 ratio, has also been employed.[46] The diarylmethanol was obtained with 97% ee, owing to the suppression of the competitive racemic pathway. A range of substituted aromatic, heteroaromatic, and aliphatic aldehydes

Table 1. Synthesis of oxazolinyl hydroxyl ferrocenes.

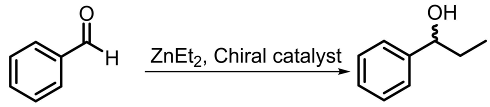
Entry	Ferrocene	Conf.	Synthetic route	Yield/%	Diastereoselectivity	Ref.
1	**3a**	*S*, *Rp*	Scheme 1	87	d.r. = 27:1	[24]
2	**3a**	*S*, *Sp*	Scheme 3	72	single dia.*	[25]
3	**3b**	*S*, *Rp*	Scheme 1	34	-	[25]
4	**3c**	*S*, *Rp*	Scheme 1	96	de > 95%	[26]
5	**3d**	*S*, *Rp*	Scheme 1	80	3:1	[26]
6	**3e**	*S*, *Rp*	Scheme 1	61	de > 95%	[26]
7	**4**	*Rp*	Scheme 2	46	-	[25]
8	**5a**	*S*, *Rp*	Scheme 1	83	de > 99%	[27]
9	**5a**	*S*, *Sp*	Scheme 1	79	de = 99%	[27]
10	**5b**	*S*, *Rp*	Scheme 1	80	de > 99%	[27]
11	**5b**	*S*, *Sp*	Scheme 3	88	de = 99%	[27]
12	**5c**	*S*, *Rp*	Scheme 1	64	single dia.*	[27]
13	**6a**	*S*, *Sp*	Scheme 1	80	d.r. > 20:1	[30]
14	**6b**	*S*, *Sp*	Scheme 1	81	d.r. > 20:1	[30]
15	**6c**	*S*, *Sp*	Scheme 1	49	d.r. > 20:1	[30]
16	**6d**	*S*, *Sp*	Scheme 1	31	d.r. > 20:1	[30]
17	**6e**	*S*, *Sp*	Scheme 1	64	d.r. > 20:1	[30]
18	**6f**	*S*, *Sp*	Scheme 1	72	d.r. > 20:1	[30]
19	**6g**	*S*, *Sp*	Scheme 1	74	d.r. > 20:1	[30]
20	**7a**	*S*, *Sp*	Scheme 6	58	-	[31]
21	**7b**	*S*, *Sp*	Scheme 6	48	-	[31]
22	**7c**	*S*, *Sp*	Scheme 6	70	-	[31]
23	**7d**	*S*, *Sp*	Scheme 6	40	-	[31]
24	**7e**	*S*, *Sp*	Scheme 6	36	-	[31]
25	**7f**	*S*, *Sp*	Scheme 6	53	-	[31]
26	**7g**	*S*, *Sp*	Scheme 6	61	-	[31]
27	**7h**	*S*, *Sp*	Scheme 6	37	-	[31]
28	**7i**	*S*, *Sp*	Scheme 6	26	-	[31]
29	**8a**	*Rp*, *Rp*	Scheme 7	62	78:22	[32]
30	**8a**	*Rp*, *Sp*	Scheme 7	17		[32]
31	**8b**	*Rp*, *Rp*	Scheme 7	69	82:18	[32]
32	**8b**	*Rp*, *Sp*	Scheme 7	15		[32]
33	**9a**	*S*, *Rp*	Scheme 8	50	-	[33]
34	**9b**	*S*, *Rp*	Scheme 8	95	-	[33]
35	**10**	*S*, *Rp*	Scheme 9	90	single dia.	[34]
36	**11a**	*S*	Scheme 10 (eq. 1)	93	-	[35]
37	**11b**	*S*	Scheme 10 (eq. 1)	82	-	[35]
38	**11c**	*S*	Scheme 10 (eq. 1)	90	-	[35]
39	**11d**	*R*	Scheme 10 (eq. 1)	73	-	[35]
40	**11e**	*S*	Scheme 10 (eq. 2)	57	-	[36]
41	**11f**	*S*	Scheme 10 (eq. 3)	66	-	[36]
42	**12a**	5*R*, 4*S*	Scheme 11 (eq. 1)	65	-	[37]
43	**12b**	5*R*, 4*S*, *Sp*	Scheme 11 (eq. 1)	57	-	[37]
44	**12c**	*R*	Scheme 12	84	-	[38]
45	**12d**	*R*	Scheme 12	92	-	[38]
46	**12e**	*S*	Scheme 13	46	-	[39]
47	**13a**	5*R*, 4*S*, 5′*R*, 4′*S*	Scheme 11 (eq. 2)	61	-	[37]
48	**13b**	*R*, *R*′	Scheme 13	43	-	[39]
49	**13c**	*S*, *S*′	Scheme 13	-	-	[39]
50	**13d**	*S*, *S*′	Scheme 13	-	-	[39]
51	**13e**	*S*, *S*′	Scheme 13	-	-	[39]
52	**24**	*S*, *Sp*	Scheme 5	68	d.r. = 11:1	[43]

* single dia. = single diastereoisomer

were examined, and the alcohols were obtained with good to high enantioselectivities.

Bolm reported the asymmetric synthesis of 1,1-disubstituted diarylmethanols using boronic acids as aryl sources and (*S*, *Rp*)-**3a** as a chiral catalyst.[47] The desired alcohols were obtained in high yields and with high enantioselectivities under harsher optimized conditions. Furthermore, substituted phenyl boronic acids were used as aryl sources to vary the structures of the chiral alcohols. A preliminary study on the effect of dimethoxy poly(ethylene glycol) (MPEG) in the catalytic reaction was also conducted. A comprehensive study on the combinations of functionalized aldehydes and substituted phenyl boronic acids was reported in 2007.[48] Both enantiomers of the alcohols could be accessed with high ee under the same catalytic conditions.

Table 2. Asymmetric addition of diethylzinc to benzaldehyde **42 a**.



42a **43a**

Entry	Catalyst	Solvent	Temp./°C	Time/h	Yield/%	*ee*/% (config.)	Ref.
1	(*S*, *Rp*)-**3 a**	toluene	0	6	83	93 (*R*)	[24]
2	(*S*, *Sp*)-**3 a**	toluene	0	59	55	35 (*R*)	[25]
3	(*S*, *Rp*)-**3 b**	toluene	0	3	99	94 (*R*)	[25]
4	(*Rp*)-**4**	toluene	0	20	97	51 (*R*)	[25]
5	(*S*, *Rp*)-**3 c**	toluene	0	24	97	83 (*R*)	[26]
6	(*S*, *Rp*)-**3 e**	toluene	0	24	95	97 (*R*)	[26]
7	(*S*, *Rp*)-**5 a**	hexane	-20	24	93	88 (*R*)	[27]
8	(*S*, *Sp*)-**5 a**	hexane	-20	24	54	82 (*R*)	[27]
9	(*S*, *Rp*)-**5 b**	hexane	-20	24	32	33 (*R*)	[27]
10	(*S*, *Sp*)-**5 b**	hexane	-20	24	11	5 (*S*)	[27]
11	(*S*, *Rp*)-**5 c**	hexane	-20	24	79	93 (*R*)	[27]
12	(*Rp*, *Rp*)-**8 a**	toluene	0	24	25	70 (*R*)	[32]
13	(*Rp*, *Rp*)-**8 b**	toluene	0	24	63	91 (*R*)	[32]
14	(*Rp*, *Sp*)-**8 a**	toluene	0	24	68	83 (*R*)	[32]
15	(*Rp*, *Sp*)-**8 b**	toluene	0	8	97	93 (*R*)	[32]
16	(*S*)-**11 a**	hexane- toluene	0	-	98	80.9 (*R*)	[35]
17	(*S*)-**11 b**	hexane- toluene	0	-	98	71.6 (*R*)	[35]
18	(*S*)-**11 c**	hexane- toluene	0	-	98	88.6 (*R*)	[35]
19	(*R*)-**11 d**	hexane- toluene	0	-	92	81.5 (*S*)	[35]
20	**12 a**	toluene	r.t.	24	76	46 (*R*)	[37]
21	**12 b**	toluene	r.t.	24	60	46 (*R*)	[37]
23	**12 c**	toluene	r.t.	24	> 95	75 (*R*)	[38]
24	**12 d**	toluene	r.t.	24	> 95	36 (*R*)	[38]
22	**13 a**	toluene	r.t.	24	80	62 (*R*)	[37]
25	**13 b**	toluene	0	48	92	79 (*R*)	[39]
26	**13 c**	toluene	0	48	93	87 (*R*)	[39]
27	**13 d**	toluene	0	48	86	45 (*R*)	[39]
28	**13 e**	toluene	0	48	86	41 (*R*)	[39]
29	**12 e**	toluene	0	48	89	20 (*R*)	[39]

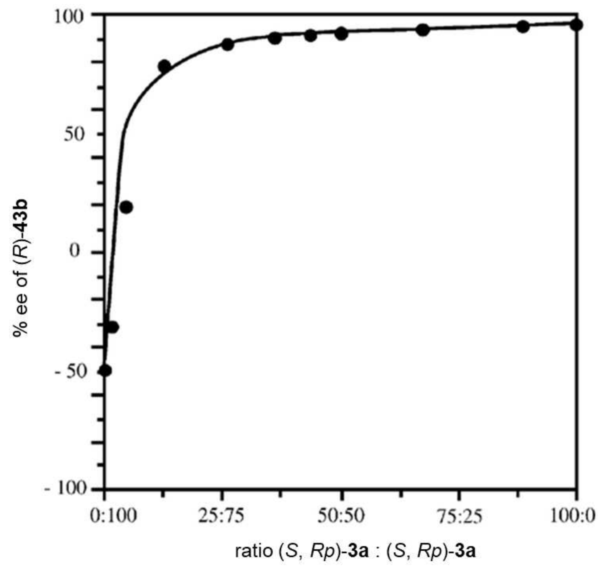


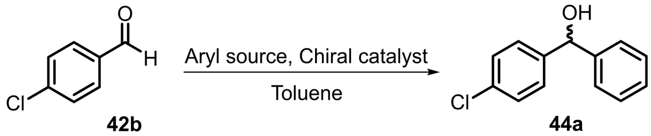
Figure 4. Nonlinear relationship between enantiomeric excesses of (*R*)-**43 b** and **3 a**. Adapted with permission from ref. [44]. Copyright 1999 American Chemical Society.

Commercially inexpensive triphenylborane has also been used as a phenyl source in the asymmetric addition catalyzed by (*S*, *Rp*)-**3 a**.[49] A survey of substrates revealed that various heteroaromatic and aliphatic aldehydes were suitable for this transformation, affording products in high yields and with good enantioselectivities (up to 97% ee). Further, the reaction was performed successfully on a gram scale, but with a slight decrease in stereoselectivity.

In an independent work, Bolm determined how MPEG enhances enantioselectivity.[50] With 10 mol% DiMPEG, remarkable improvements in enantioselectivity were observed when either $ZnPh_2$/$ZnEt_2$ or BPh_3 was used as a phenyl source. It was presumed that the presence of MPEG as an additive deactivated the nonenantioselective process, leading to the improvement. In view of this, the catalyst system was scaled up to a multigram scale and arylboronic acids were used as the aryl sources.[51] Enantioenriched diarylmethanols were obtained (up to 96% ee) and the ferrocene catalyst, (*S*, *Rp*)-**3 a**, could be recovered and reused without loss of catalytic activity or selectivity.

The Guiry research group reported enantioselective phenyl transfer catalyzed by *gem*-disubstituted ferrocene **5**.[27] Steric repulsion was assumed to be the essential factor controlling this asymmetric process. The yields were similar to those

Table 3. Asymmetric phenyl transfer to *p*-chloro-benzaldehyde **42 b**.



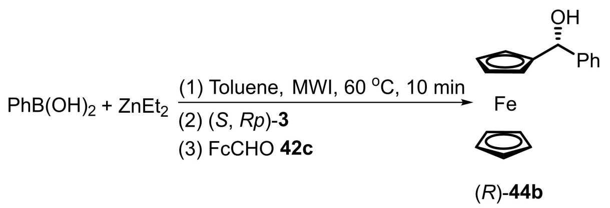
Entry	Catalyst	Aryl Source	Temp./°C	Time/h	Yield/%	*ee*/% (config.)	Ref.
1	(*S*, *Rp*)-**3 a**	$ZnPh_2$	0	15	99	88 (*R*)	[45]
2	(*S*, *Rp*)-**3 b**	$ZnPh_2$	0	13	99	88 (*R*)	[45]
3	(*S*, *Rp*)-**3 a**	$ZnPh_2/ZnEt_2$	10	-	86	97 (*R*)	[46]
4	(*S*, *Rp*)-**3 a**	$PhB(OH)_2/ZnEt_2$	60	12	95	92 (*R*)	[47]
5*	(*S*, *Rp*)-**3 a**	$PhB(OH)_2/ZnEt_2$	10	12	93	97 (*R*)	[48]
6	(*S*, *Rp*)-**3 a**	$BPh_3/ZnEt_2$	10	12	98	97 (*R*)	[49]
7	(*S*, *Rp*)-**5 a**	$ZnPh_2/ZnEt_2$	10	24	82	92 (*R*)	[27]
8	(*S*, *Sp*)-**5 a**	$ZnPh_2/ZnEt_2$	10	24	81	68 (*S*)	[27]
9	(*S*, *Rp*)-**5 b**	$ZnPh_2/ZnEt_2$	10	24	48	85 (*R*)	[27]
10	(*S*, *Sp*)-**5 b**	$ZnPh_2/ZnEt_2$	10	24	32	30 (*S*)	[27]
11	(*S*, *Rp*)-**5 c**	$ZnPh_2/ZnEt_2$	10	24	93	-	[27]
12	(*S*, *Sp*)-**7 a**	$ZnPh_2/ZnEt_2$	10	12	40	30 (*R*)	[31]
13	(*S*, *Sp*)-**7 b**	$ZnPh_2/ZnEt_2$	10	12	82	91 (*R*)	[31]
14	(*S*, *Sp*)-**7 c**	$ZnPh_2/ZnEt_2$	10	12	84	89 (*R*)	[31]
15	(*S*, *Sp*)-**7 c**	$BPh_3/ZnEt_2$	10	12	73	88 (*R*)	[31]
16*	(*S*, *Sp*)-**7 c**	$PhB(OH)_2/ZnEt_2$	60	12	67	83 (*R*)	[31]
17	(*S*, *Sp*)-**7 d**	$ZnPh_2/ZnEt_2$	10	12	82	87 (*R*)	[31]
18	(*S*, *Sp*)-**7 e**	$ZnPh_2/ZnEt_2$	10	12	76	85 (*R*)	[31]
19	(*S*, *Sp*)-**7 f**	$ZnPh_2/ZnEt_2$	10	12	87	78 (*R*)	[31]
20	(*S*, *Sp*)-**7 g**	$ZnPh_2/ZnEt_2$	10	12	73	76 (*R*)	[31]
21	(*S*, *Sp*)-**7 h**	$ZnPh_2/ZnEt_2$	10	12	85	63 (*R*)	[31]
22	(*S*, *Sp*)-**7 i**	$ZnPh_2/ZnEt_2$	10	12	n.d.	58 (*R*)	[31]

* 10 mol% DiMPEG (Mw = 2000 g/mol) as an additive.

obtained by Bolm and the enantioselectivities were slightly lower, and this could be ascribed to the greater steric bulkiness of the substituents on ferrocene **5**. This presumption was based on the results obtained using (*S*, *Rp*)-**5 c**, which afforded high enantioselectivities in ethyl transfer, but provided only racemic diarylmethanols in phenyl transfer.

Planar chiral oxazolinyl silanols **7** were all applied as catalysts in asymmetric phenyl transfer to benzaldehyde by Bolm and coworkers.[31] High yields and enantioselectivities (up to 91% ee) were obtained after organosilanols and phenyl sources were surveyed.

Assisted by microwave irradiation (MWI), Šebesta demonstrated quick enantioselective synthesis of aryl(ferrocenyl) methanols in good yields and with good enantiomeric ratios (Scheme 15).[52] The only drawback of this protocol was a slightly lower enantioselectivity than that in the asymmetric reduction of prochiral ferrocenyl ketones.



Scheme 15. Enantioselective addition of arylboronic acids to ferrocenecarboxaldehyde.

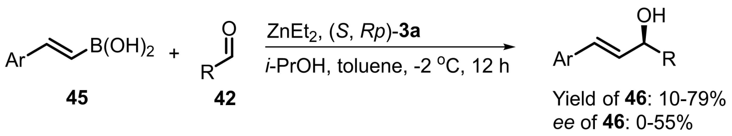
4.3 Recycling of polymer-supported ferrocenyl catalysts in asymmetric additions to aldehydes

Polymer-supported ferrocene catalysts were first proposed for asymmetric additions to aldehydes by Bolm.[33] Enantioselective phenyl transfer to 4-chlorobenzaldehyde was selected as the model reaction for the recycling of ferrocene catalysts **9 b**. Quantitative yield and 97% ee were obtained using **9 b** and $ZnPh_2/ZnEt_2$ as the phenyl source. Moreover, MeO-PEG-supported **9 b** could be reused in five catalytic cycles without apparent loss of enantioselectivity (Table 4).

The latest example of a polymer-tagged catalyst was developed by Zhao and Liu, who applied ferrocene **10** to asymmetric ethyl transfer to several aromatic acceptors.[34] In most cases, the addition products were obtained in good yields and with moderate ee values. In the recycling experiments, the polymer catalyst retained activity for three cycles, while marked

Table 4. Recycling experiments of enantioselective phenyl transfer to *p*-chloro-benzaldehyde **42 b**.

Entry	Cycle	Yield of **44 a**/%	*ee* of **44 a**/% (config.)
1	1	80	97 (*R*)
2	2	75	96 (*R*)
3	3	81	95 (*R*)
4	4	97	95 (*R*)
5	5	80	95 (*R*)



Scheme 16. Asymmetric alkenylzinc addition to aldehydes.

decreases in yield and ee were observed in the fourth run (Table 5).

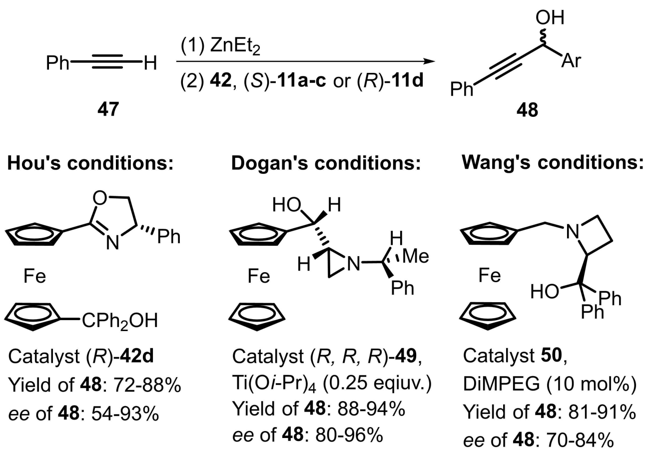
4.4 Miscellaneous examples

Enantioselective addition of alkenylzinc to aldehydes is a powerful approach for obtaining chiral allyl alcohols. Bolm employed alkenylboronic acids as alkenyl sources, which were reacted with $ZnEt_2$. Using (*S*, *Rp*)-**3 a** as a chiral catalyst, desired asymmetric allyl alcohols **46** were obtained in moderate yields and enantiomeric excess (Scheme 16).[53]

Zinc-mediated enantioselective alkynylation has been widely discussed in recent years (Scheme 17). Enantioselective alkynylation catalyzed by oxazolinyl hydroxyl ferrocenes was described by Hou and coworkers in 2004.[36] Reactions between phenylacetylene and aldehydes with 1,1'-substituted oxazolinyl hydroxyl ferrocene (*R*)-**11 d** as the catalyst gave good yields and enantioselectivities, with 90% ee. Notably, subsequent examples of asymmetric alkynylation with ferrocene-based catalysts were reported by Dogan[54] and Wang[55] independently in 2007. Propargylic alcohols **48** were obtained with up to 96% ee using aziridinylmethanol ferrocenes **49** in the presence of 0.25 equiv. of Ti(O*i*-Pr)$_4$.

Table 5. Recycling experiments of asymmetric ethyl transfer to benzaldehyde **42 a**.

Entry	Cycle	Yield of 43 a/%	*ee* of 43 a/% (config.)
1	-	>99	88 (*R*)
2	1	97	83 (*R*)
3	2	97	81 (*R*)
4	3	80	62 (*R*)



Scheme 17. Enantioselective alkynylation of aldehydes **42** using hydroxyl ferrocenes.

5. Conclusion and perspectives

In this account, we intensively surveyed the chiral oxazolinyl hydroxyl ferrocenes that have been published in the literatures, particularly the recent advances in diastereoselective synthesis and enantioselective applications of ferrocene derivatives. The diversity in the structures of the chiral ferrocenes has broadened in the past decades, because of practical functionalization of aromatic Cp ring. Correspondingly, elegant examples of methods on synthesis of ferrocenes for each pattern have been developed by several research groups. These *N*,*O*-bidentate ferrocenes have predominantly been served as chiral catalysts in zinc-mediated enantioselective nucleophilic additions to aldehydes, and proved to be competitive catalysts employed in enantioselective alkyl and aryl transfers to aldehydes.

While the achievements on the enlisted ferrocene-based catalysts are notable, the application scopes of the enlisted ferrocene-based catalysts are restricted to enantioselective nucleophilic additions. First, unlike the *N*,*P*-chelated transition-metal catalysts, most of *N*,*O*-bidentate catalysts should be activated by strong Lewis bases, such as $ZnMe_2$ and $ZnEt_2$. There, the resulting chiral complexes are limited to the zinc-mediated transformation. Second, synthesis of chiral unsaturated alcohols using soft nucleophiles (alkenylboronic acids, phenylacetylenes) by the reported chiral oxazolinyl hydroxyl ferrocenes remains a challenge. In fact, ProPhenol compounds,[21] that used as bifunctional catalysts activating two reagents simultaneously, have provided sufficient results in asymmetric alkyne addition. Third, the enantioselective processes of chiral alcoholic compounds by nucleophilic additions are hampered by utilization of moisture sensitive organometallic reagents under harsh reaction conditions. In comparison, asymmetric reduction of unsymmetrical ketones under mild condition is illustrated to be efficient and atom-economical approach to access complicated unsaturated alcohols.

Currently, chiral transition metal catalysts and ligands are being designed according to changes in the target chiral intermediates and compounds. As far as chiral ferrocene ligands/catalysts are concerned, there are several important developmental trends:

(1) Stereodivergent synthesis

In a study on disubstituted ferrocenes applied in asymmetric nucleophilic addition, Guiry found that the absolute configurations of stereogenic elements attached to the ferrocene framework directly influenced the configurations of the corresponding chiral alcohols.[27] Very recently, Richards discussed the impacts of all diastereomers of oxazolinyl phosphine ferrocenes (FOXAPs) on palladium-catalyzed allylic alkylation. The results depend strongly on the stereodivergent synthesis of chiral ferrocenes.[56] We believe that a greater number of step-economic strategies for stereodivergent synthesis of planar chiral ferrocene derivatives will be explored in the near future.

(2) Task-specific ligands

The absolute configurations of chiral compounds with multiple stereogenic centers directly influence their biological and pharmaceutical activities. Therefore, efficient enantioselective synthesis of specific diastereomers has received more attention in recent years.[57–59] Stereodivergent strategies involving task-specific ligands has emerged. Hou reported that tuning the electric properties of the phosphorus atoms on FOXAP ligands leads to chiral pyrrolidines via 1,3-dipolar cycloaddition in *exo* and *endo* conformations, and each diastereomer was obtained with high selectivity.[60] Asymmetric stereodivergent synthesis promoted by ferrocene-type ligands was also developed by Cossío,[61] Wang,[62] and Fukuzawa.[63] Because of these encouraging advances, the exploration of diastereomer-specific ferrocenyl ligands is emerging.

(3) Sustainable transition metal catalysts

Because of the demand for economical and environmentally friendly catalytic processes, recyclable catalysts have been a great concern in asymmetric transformations. In this review, we discussed immobilized PEG groups and the recycling procedures for ferrocene catalysts independently reported by Bolm and Zhao. The derived polymer-tagged catalysts could be reused at least three times without any obvious loss of activity or enantioselectivity. In addition, functionalized ferrocene ligands with hydrophilic groups, such as ionic liquids, have been reported by our group.[64–65] The modified ferrocenes also exhibit robustness during the recycling process.

The urgent demand for higher enantioselectivity has prompted researchers to explore structures with oxazolinyl ferrocene cores. We hope that this review will draw attention to these fascinating ferrocenyl catalysts and stimulate further research in this attractive field.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords: asymmetric catalysis · nucleophilic addition · oxazolinyl ferrocene · secondary alcohols.

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